ECE 476 Formula Sheet

Chukwugozie Okeke

Rectangular Waveguides

Basic Waveguide Parameters

$$k = \omega\sqrt{\mu\epsilon} = \frac{\omega\mu}{\eta} \qquad k_x = \frac{m\pi}{a} \qquad k_y = \frac{n\pi}{b} \qquad k_z = \beta$$

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\omega^2\mu\epsilon - k_c^2} \qquad k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

Other Wave Parameters

$$\beta = \frac{2\pi}{\lambda_g} \qquad k_c = \frac{2\pi}{\lambda_c} \qquad k = \frac{2\pi}{\lambda_0} \qquad \lambda_g = \frac{\lambda_0}{\sqrt{1-(\lambda_0/\lambda_c)^2}} = \frac{\lambda_0}{\cos\theta}$$

$$Z_{TE} = \frac{\omega\mu}{\beta} = \frac{k\eta}{\beta} \qquad Z_{TM} = \frac{\beta}{\omega\epsilon} = \frac{\beta\eta}{k} \qquad BW = f_H - f_L \xrightarrow{\%} \frac{2(f_H - f_L)}{(f_H + f_L)}$$

$$v_{ph} = \frac{\omega}{\beta} = \frac{v_0}{\cos\theta} \qquad v_g = v_0\cos\theta \qquad v_g v_{ph} = c^2 \qquad \cos\theta = \beta/k$$

TE_{mn} *Mode Field Equations* ***(Dominant: TE_{10})***

$$H_z = A_{mn}\cos\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$E_x = \frac{j\omega\mu n\pi}{k_c^2 b}A_{mn}\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$E_y = \frac{-j\omega\mu m\pi}{k_c^2 a}A_{mn}\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$H_x = \frac{j\beta m\pi}{k_c^2 a}A_{mn}\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$H_y = \frac{j\omega n\pi}{k_c^2 b}A_{mn}\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

TM_{mn} *Mode Field Equations* ***(Dominant: TM_{11})***

$$E_z = B_{mn}\sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$E_x = \frac{-j\beta m\pi}{k_c^2 a}B_{mn}\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$E_y = \frac{-j\beta n\pi}{k_c^2 b}B_{mn}\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$H_x = \frac{j\omega\epsilon n\pi}{k_c^2 b}B_{mn}\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

$$H_y = \frac{-j\omega\epsilon m\pi}{k_c^2 a}B_{mn}\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{-j\beta z}$$

TE_{10} *Power Characteristics*

$$P_{T10} = \frac{\omega\mu a^3|A_{10}|^2 b}{4\pi^2}\Re\{\beta\} \qquad P_{L10} = R_s|A_{10}|^2\left[b + \frac{a}{2} + \frac{\beta^2 a^3}{2\pi^2}\right]$$

$$\alpha = \frac{P_L}{2P_T} = \frac{2\pi^2 R_s\left(b + a/2 + \beta^2 a^2/2\pi^2\right)}{\omega\mu a^3 b\beta}\ [\text{Np/m}] \qquad 1\text{Np} = 8.686\ \text{dB}$$

Rectangular Waveguide Cavities (Resonators)

Waveguide Cavity Parameters

$$\beta_{mn} = \frac{l\pi}{d} \qquad \omega = 2\pi f = \frac{1}{\sqrt{\mu\epsilon_0\epsilon_r}}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{l\pi}{d}\right)^2}$$

$$k^2 = k_c^2 + \beta_{mn}^2$$

TE_{10l} *Mode Field Equations* ***(Dominant: TE_{101})***

$$E_y = E_0\sin\left(\frac{\pi x}{a}\right)sin\left(\frac{l\pi z}{d}\right) \qquad H_x = \frac{-jE_0}{Z_{TE}}\sin\left(\frac{\pi x}{a}\right)cos\left(\frac{l\pi z}{d}\right)$$

$$H_z = \frac{j\pi E_0}{k\eta a}\cos\left(\frac{\pi x}{a}\right)sin\left(\frac{l\pi z}{d}\right)$$

TE_{10l} *Mode Power Characteristics*

$$W_{Sto.} = W_E + W_H \qquad W_E = W_H = \frac{\epsilon abd}{16}E_0^2 \qquad Q_c = \frac{f_r}{BW} = \frac{\omega W_{Sto.}}{P_L}$$

$$P_L = \frac{R_s E_0^2\lambda^2}{8\eta^2}\left[\frac{l^2 ab}{d^2} + \frac{bd}{a^2} + \frac{l^2 a}{2d} + \frac{d}{2a}\right] \qquad P_D = \frac{abd\omega\epsilon''|E_0|^2}{8}$$

$$Q_c = \frac{(kad)^3 b\eta}{2\pi^2 R_s}\frac{1}{2l^2a^3b + 2bd^3 + l^2a^3d + ad^3}$$

$$\frac{1}{Q_T} = \frac{1}{Q_c} + \frac{1}{Q_d} \qquad Q_d = \frac{\omega W_{Sto.}}{P_d} = \frac{\epsilon'}{\epsilon''} = \frac{1}{tan(\delta)} \qquad \epsilon = \epsilon' + j\epsilon''$$

Circular Waveguides

Basic Circular Waveguide Parameters

$$k_{c_{nm}} = \rho'_{n,m}/a \qquad f_c = \frac{k_c c}{2\pi} \qquad \textbf{Note}: TE_{0m} \rightarrow \text{loss dec. w/ freq.}$$

n	$\rho'_{n,1}$	$\rho'_{n,2}$	$\rho'_{n,3}$
0	3.832	7.016	10.174
1	1.841	5.331	8.536
2	3.054	6.706	9.970

TE_{nm} *Mode Field Equations* ***(Dominant: TE_{11})***

$$H_z = (A\sin(n\phi) + B\cos(n\phi))\,J_n(k_c r)e^{-j\beta z} \qquad \alpha_c = \frac{R_s}{ak\eta\beta}\left(k_c^2 + \frac{k^2}{(\rho'^2_{11} - 1)}\right)$$

$$E_r = \frac{-j\omega\mu n}{k_c^2 r}(A\sin(n\phi) - B\cos(n\phi))\,J_n(k_c r)e^{-j\beta z}$$

$$E_\phi = \frac{j\omega\mu}{k_c^2}(A\sin(n\phi) + B\cos(n\phi))\,J'_n(k_c r)e^{-j\beta z}$$

$$H_r = \frac{-j\beta}{k_c}(A\sin(n\phi) + B\cos(n\phi))\,J'_n(k_c r)e^{-j\beta z}$$

$$H_\phi = \frac{-j\beta n}{k_c^2 r}(A\sin(n\phi) - B\cos(n\phi))\,J_n(k_c r)e^{-j\beta z}$$

TE_{11} *Power Characteristics*

$$P_0 = \frac{\pi\omega|A|^2\Re\{\beta\}}{4k_c^4}(\rho_{11}^2 - 1)J_1^2(k_c a) \qquad P_L = \frac{\pi|A|^2 R_s a}{2}\left(1 + \frac{\beta^2}{k_c^4 a^2}\right)J_1^2(k_c a)$$

Other Important Formulas

Power Flow and Storage (General)

$$\bar{J}_s = \hat{n}\times(\bar{H}_2 - \bar{H}_1) \qquad \bar{S} = \bar{E}\times\bar{H}^* \qquad P_T = \frac{1}{2}\oint_S \bar{S}\cdot d\bar{s} \qquad R_s = \sqrt{\frac{\omega\mu}{2\sigma}}$$

$$P_L = \frac{R_S}{2}\int|\bar{J}_s|^2 dl \qquad \alpha = \frac{P_L}{2P_T} \qquad P(z) = P_0 e^{-2\alpha z} \qquad W_e = \frac{\epsilon}{4}\int_V \bar{E}\cdot\bar{E}^* dv$$

$$W_m = \frac{\mu}{4}\int_V \bar{H}\cdot\bar{H}^* dv$$

Complex Wavenumber (Plane Waves)

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta \qquad \alpha,\,\beta = \omega\sqrt{\frac{\mu\epsilon}{2}\left(\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} \mp 1\right)}$$

Antennas

Antenna Parameters

$$D(\theta,\phi)=\frac{U(\theta,\phi)}{U_0} \qquad D_{max}=\frac{U_{max}}{U_0}=\frac{4\pi U_{max}}{P_{rad}} \qquad R_{RNF}=\sqrt{\frac{D^3}{\lambda}}$$

$$R_{FF}=\frac{2D^2}{\lambda} \qquad G=e_{cd}D=\frac{4\pi U(\theta,\phi)}{P_{in}} \qquad \Omega_A=\frac{P_{rad}}{U_{max}}$$

Antenna Power

$$W_{Rad}=\frac{1}{2}\Re\{E\times H^*\} \qquad U=r^2W_{rad} \qquad P_{rad}=\int_\Omega U(\theta,\phi)d\Omega$$

Polarization ***(Always Check Normalization)***

$$\rho_{lin}=\frac{a\hat{x}+b\hat{y}}{\sqrt{a^2+b^2}} \qquad \tau=\tan^{-1}\left(\frac{E_y}{E_x}\right)$$

$-\hat{z}$: $\rho_t^{RH}=\frac{\hat{x}+j\hat{y}}{\sqrt{2}}$ $\rho_r^{RH}=\frac{\hat{x}-j\hat{y}}{\sqrt{2}}$ $\rho_t^{LH}=\frac{\hat{x}-j\hat{y}}{\sqrt{2}}$ $\rho_r^{LH}=\frac{\hat{x}+j\hat{y}}{\sqrt{2}}$

Friis Transmission Equation

$$e_r=1-|\Gamma|^2 \qquad |\Gamma|=\left|\frac{Z_{in}-Z_c}{Z_{in}+Z_c}\right| \qquad VSWR=\frac{V_i+V_r}{V_i-V_r}=\frac{1+|\Gamma|}{1-|\Gamma|}$$

$$|\Gamma|=\frac{VSWR-1}{VSWR+1} \quad A_e=\frac{\lambda^2}{4\pi}e_{cd}D=\frac{\lambda^2}{4\pi}G \quad P_r=A_eW_i \quad PLF=|\hat{\rho}_t\cdot\hat{\rho}_r|^2$$

$$P_r=P_t\left(\frac{\lambda}{4\pi R}\right)^2 e_{cdt}e_{cdr}D_t(\theta,\phi)D_r(\theta,\phi)\left(1-|\Gamma_t|^2\right)\left(1-|\Gamma_t|^2\right)|\hat{\rho}_t\cdot\hat{\rho}_r|^2$$

$$P_r=P_t\left(\frac{\lambda}{4\pi R}\right)^2 G_t(\theta_t,\phi_t)G_r(\theta_r,\phi_r)\left(1-|\Gamma_t|^2\right)\left(1-|\Gamma_t|^2\right)|\hat{\rho}_t\cdot\hat{\rho}_r|^2$$

Antenna Thevenin Equivalents

$$Z_A=R_r+R_L+jX_A \qquad Z_g=R_g+jX_g \qquad P=\frac{1}{2}|I_g|^2R$$

Match Cond. $Z_g=Z_A^*\rightarrow R_r+R_L=R_g, X_A=-X_g$

Radiation and Radiated Fields

$$\bar{A}=\frac{\mu}{4\pi}\int_V \bar{J}\frac{e^{-jkR}}{R}dv' \qquad \bar{F}=\frac{\epsilon}{4\pi}\int_V \bar{M}\frac{e^{-jkR}}{R}dv'$$

$$\bar{H}_A=\frac{1}{\mu}\nabla\times\bar{A} \qquad E_A=-j\omega\bar{A}-j\frac{1}{\omega\mu\epsilon}\nabla(\nabla\cdot\bar{A}) \qquad E_A=-\frac{j}{\omega\epsilon}\nabla\times\bar{H}_A$$

$$\bar{E}_F=-\frac{1}{\epsilon}\nabla\times\bar{F} \qquad H_F=-j\omega\bar{F}-j\frac{1}{\omega\mu\epsilon}\nabla(\nabla\cdot\bar{F}) \qquad H_F=\frac{j}{\omega\mu}\nabla\times\bar{E}_F$$

Dipole Antennas

Infinitesimal Dipole ($L<\lambda/50$, $D=3/2$)

$$\bar{A}=\frac{\mu}{4\pi}I_0L\frac{e^{-jkr}}{r}\hat{z} \qquad \bar{E}_\theta=\frac{j\eta kI_0L}{4\pi r}e^{-jkr}\sin(\theta) \qquad \bar{H}_\phi=\frac{jkI_0L}{4\pi r}e^{-jkr}\sin(\theta)$$

$$W_{rad}=\frac{\eta}{2}\left(\frac{kI_0L}{4\pi R}\right)^2\sin^2(\theta) \qquad P_{rad}=\frac{\pi\eta}{3}\left(\frac{I_0L}{\lambda}\right)^2 \qquad R_{rad}=\frac{2\pi\eta}{3}\left(\frac{L}{\lambda}\right)^2$$

Small Dipole ($\lambda/50<L<\lambda/10$)

$$\bar{A}=\frac{\mu}{8\pi}I_0L\frac{e^{-jkr}}{r}\hat{z} \qquad \bar{E}_\theta=\frac{j\eta kI_0L}{8\pi r}e^{-jkr}\sin(\theta) \qquad \bar{H}_\phi=\frac{jkI_0L}{8\pi r}e^{-jkr}\sin(\theta)$$

$$R_{rad}=20\pi^2\left(\frac{L}{\lambda}\right)^2$$

Finite Length Dipole ($L>\lambda/10$)

$$R_{in}=\frac{R_{rad}}{\sin^2(kL/2)} \qquad I(z)=\hat{z}I_0\begin{cases}\sin\left[k\left(\frac{L}{2}-z\right)\right], & 0\le z\le L/2\\ \sin\left[k\left(\frac{L}{2}+z\right)\right], & -L/2\le z\le 0\end{cases}$$

$$E_\theta=\eta H_\phi=j\eta\frac{I_0e^{-jkr}}{2\pi r}\left[\frac{\cos\left(k\frac{L}{2}\cos(\theta)\right)-\cos\left(k\frac{L}{2}\right)}{\sin(\theta)}\right]$$

$$G=\frac{kL}{2} \qquad R_{in}=\begin{cases}20G^2, & 0\le L\le\lambda/4\\ 24.7G^{2.5}, & \lambda/4\le L\le\lambda/2\\ 11.14G^{4.17}, & \lambda/2\le L\le 0.637\lambda\end{cases}$$

Half-Wave Dipole ($L=\lambda/2$)

$$U\approx\eta\frac{|I_0|^2}{8\pi^2}\sin^3\theta \quad P_{rad}=\eta\frac{|I_0|^2}{8\pi}2.435 \quad D_0=1.643=2.1\text{dB} \quad R_{rad}=73\Omega$$

Antenna Arrays

Two Element Array

$$AF=2\cos\left(\frac{1}{2}(kdcos(\theta)+\beta)\right) \qquad (AF)_n=2\cos\left(\frac{kd}{2}cos(\theta)+\beta\right)$$

N-Element Linear Array ***(Progressive Phase, Same Amplitude)***

$$AF=\frac{sin(N\phi/2)}{sin(\phi/2)}\approx\frac{sin(N\phi/2)}{\phi/2} \qquad (AF)_n=\frac{sin(N\phi/2)}{Nsin(\phi/2)}\approx\frac{sin(N\phi/2)}{N\phi/2}$$

$\phi=kd\cos(\theta)+\beta$ B.S. $(\theta=\pi/2)\rightarrow\beta=0$ E.F. $(\theta=0)\rightarrow\beta=-kd$

NxM-Element Rectangular Array ***(Progressive Phase, Same Amplitude)***

$$(AF)_T=(AF)_v(AF)_h=\frac{1}{MN}\frac{sin(N\phi/2)}{sin(\phi/2)}\frac{sin(M\xi/2)}{sin(\xi/2)} \qquad \xi=kd\cos(\theta)+\rho$$

Microstrip Patch Antennas

Design Procedure ***(Specify ϵ_r, f_r, h, find W, L)***

$$W=\frac{1}{2f_r\sqrt{\mu_0\epsilon_0}}\sqrt{\frac{2}{\epsilon_r+1}} \qquad \epsilon_{reff}=\frac{\epsilon_r+1}{2}+\frac{\epsilon_r-1}{2}\left[1+12\frac{h}{W}\right]^{-1/2}$$

$$\frac{\Delta L}{h}=0.412\frac{(\epsilon_{reff}+0.3)\left(\frac{W}{h}+0.264\right)}{(\epsilon_{reff}-0.258)\left(\frac{W}{h}+0.8\right)} \qquad L_p=\frac{1}{2f_r\sqrt{\epsilon_{reff}}\sqrt{\mu_0\epsilon_0}}-2\Delta L$$

$$Z_{in}(y_0)=Z_{in}(0)\cos^2\left(\frac{\pi}{L}y_0\right) \qquad Z_c=\frac{120\pi/\sqrt{\epsilon_{reff}}}{\frac{W_0}{h}+1.393+0.667\ln\left(\frac{W_0}{h}+1.444\right)}$$

Horn Antennas

Design Procedure ***(Specify a, b, λ, G_0, find $\chi, a_1, b_1, \mathbf{p_{e/h}}$)***

$$D_p=\frac{\pi\lambda^2}{32ab}D_ED_H \qquad a_1=\sqrt{3\lambda\rho_2} \qquad b_1=\sqrt{2\lambda\rho_1}$$

$$D_E=\frac{a}{\lambda}\frac{G_E}{\sqrt{\frac{50}{\rho_e/\lambda}}} \qquad G_E=\frac{32}{\pi}B \qquad B=\frac{b_1}{\lambda}\sqrt{\frac{50}{\rho_e/\lambda}} \qquad a\rightarrow b\ e\rightarrow h\ B\rightarrow A$$

$$\delta(y')=-\rho_1+\sqrt{\rho_1^2+y'^2}\approx\frac{1}{2}\left(\frac{y'^2}{\rho_1}\right) \quad \delta(x')=-\rho_2+\sqrt{\rho_2^2+x'^2}\approx\frac{1}{2}\left(\frac{x'^2}{\rho_2}\right)$$

$$\frac{p_e}{b_1-b}=\frac{\rho_1}{b_1} \qquad \rho_1^2=\rho_e^2-\left(\frac{b_1}{2}\right)^2 \qquad \frac{p_h}{a_1-a}=\frac{\rho_2}{a_1} \qquad \rho_2^2=\rho_h^2-\left(\frac{a_1}{2}\right)^2$$

$$G_0=\frac{1}{2}\frac{4\pi}{\lambda^2}a_1b_1 \qquad \text{Long Horn}\rightarrow\rho_e\approx\rho_1,\ \rho_h\approx\rho_2$$

$$\chi=\frac{\rho_e}{\lambda} \qquad \left(\sqrt{2\chi}-\frac{b}{\lambda}\right)^2(2\chi-1)=\left(\frac{G_0}{2\pi}\sqrt{\frac{3}{2\pi}}\frac{1}{\sqrt{\chi}}-\frac{a}{\lambda}\right)^2\left(\frac{G_0^2}{6\pi^3}\frac{1}{\chi}-1\right)$$

$$\chi_1=\frac{G_0}{2\pi\sqrt{2\pi}} \qquad \rho_e=\chi\lambda \qquad \rho_h=\frac{G_0^2}{8\pi^3}\frac{1}{\chi}\lambda \qquad a_1=\frac{G_0}{2\pi}\sqrt{\frac{3}{2\pi\chi}}\lambda \qquad b_1=\sqrt{2\chi}\lambda$$

Other Antenna Identities

$$dBd=dBi-2.1 \quad \bar{J}=\hat{n}\times\bar{H} \quad \bar{M}=-\hat{n}\times\bar{E} \quad D\approx\frac{41253}{\theta_E\theta_H} \quad G\approx\frac{30000}{\theta_E\theta_H}$$